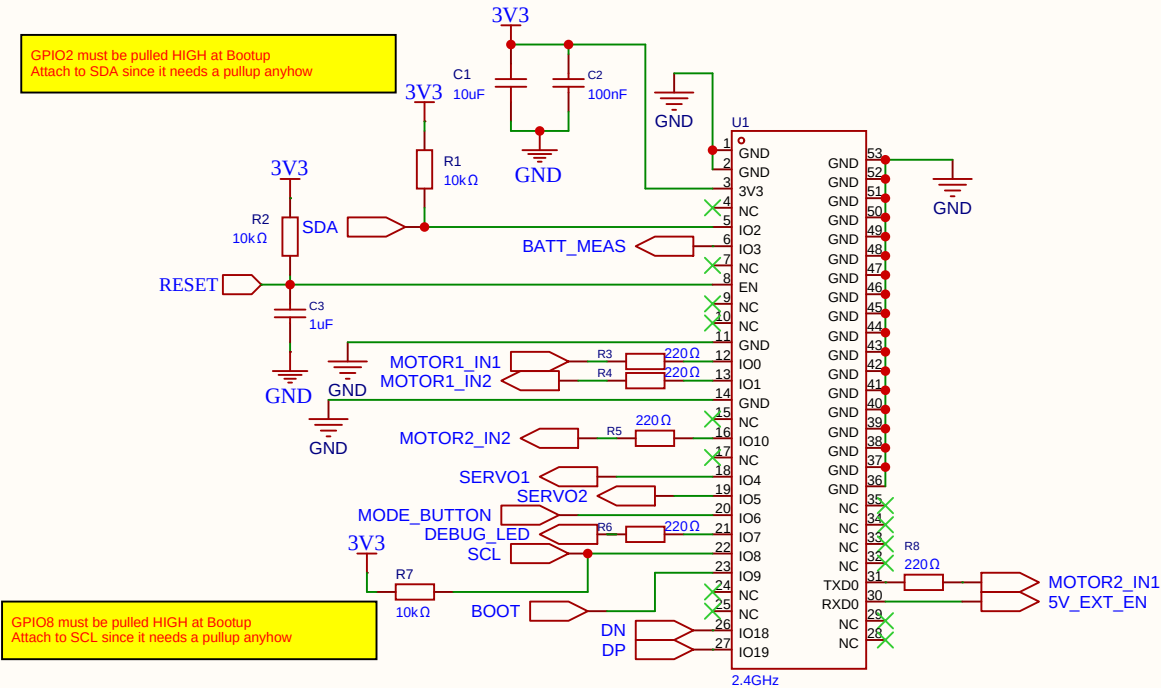
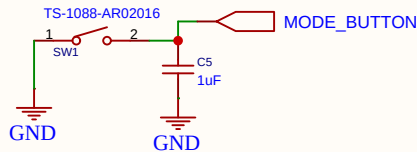




Manual Push Button

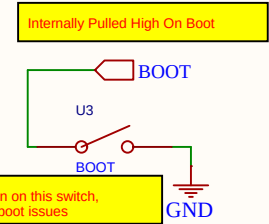
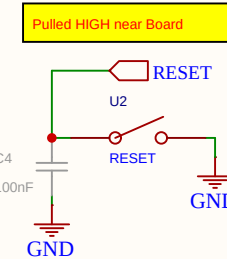


USB Boot and Reset

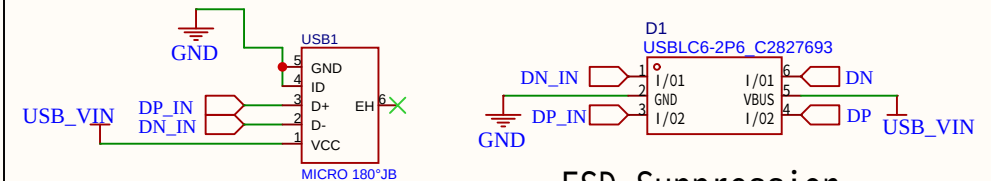
Table 6: Chip Boot Mode Control

Boot Mode	GPIO2 ^a	GPIO8	GPIO9
Default configuration	~ (Floating)	~ (Floating)	1 (Pull-up)
SPI Boot (default)	1	Any value	1
Joint Download Boot ^b	1	1	0

^a GPIO2 actually does not determine SPI Boot and Joint Download Boot mode, but it is recommended to pull this pin up due to glitches.



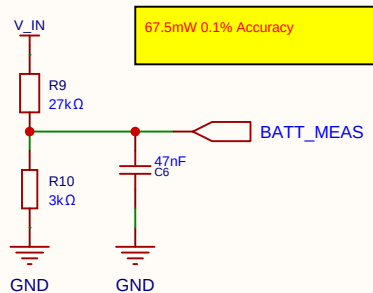
USB-Micro Connection



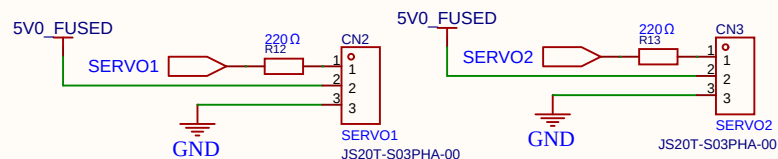
ESD Suppression

Schematic	Schematic1_1			Update Date	2025-07-31
Page	MCU Connections			Create Date	2025-02-17
Drawn	EasyEDA	Generic_Robot_Controller_ver2			
Reviewed	EasyEDA				
		VER	SIZE	PAGE	1 OF 5
EasyEDA		V0.1	A4	EasyEDA.com	

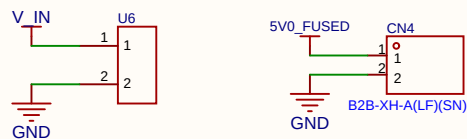
Battery Measurement



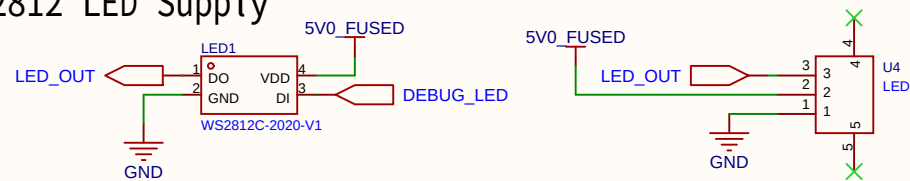
Servo Outputs



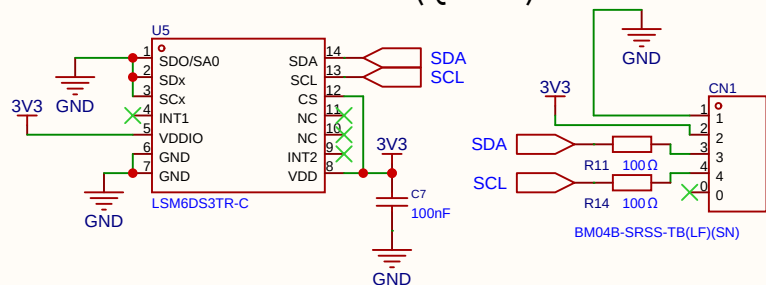
Power Out



WS2812 LED Supply



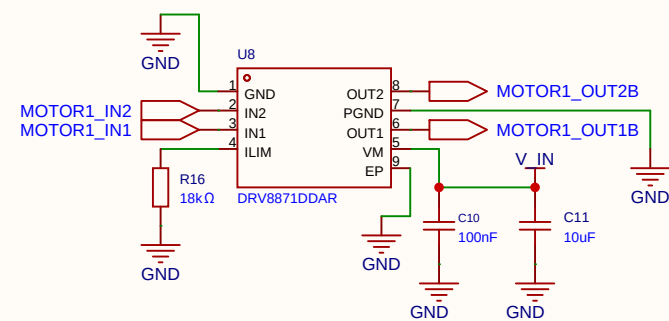
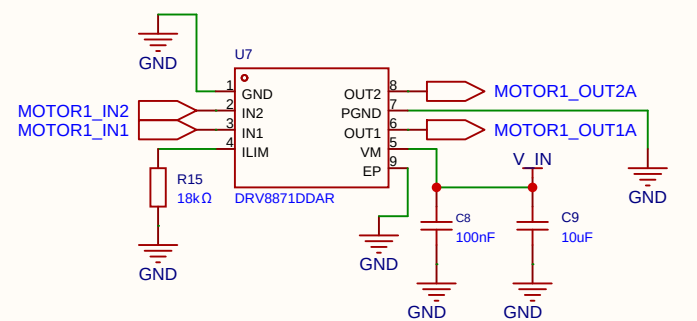
IMU and I2C Breakout (QWIIC)



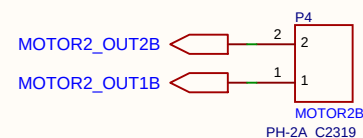
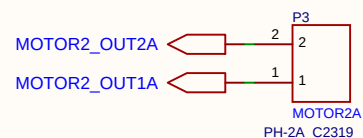
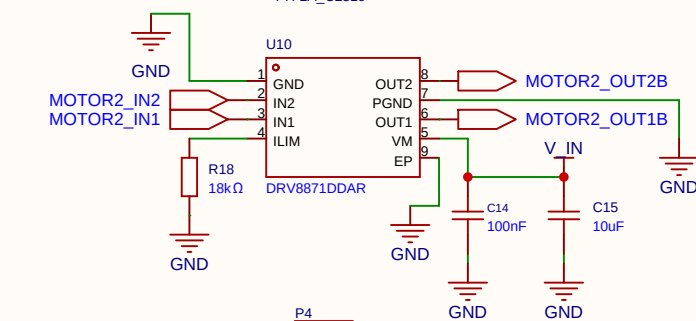
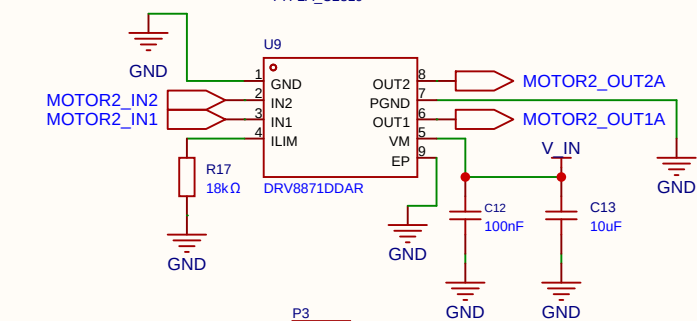
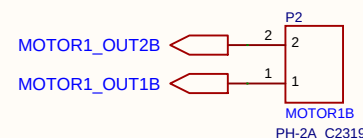
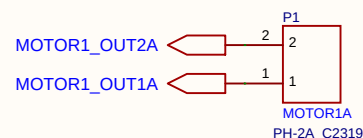
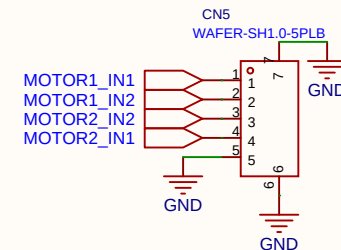
Schematic	Schematic1_1		Update Date	2025-08-07
			Create Date	2025-02-17
Page	Peripherals		Part Number	JLCPCB-002
Drawn	EasyEDA	Generic_Robot_Controller_ver2		
Reviewed	EasyEDA			
		VER	SIZE	PAGE 2 OF 5
EasyEDA		V0.1	A4	EasyEDA.com

Motor Drivers

Current Shutoff at 3.6 Amp
Voltage Input: 6.5V to 30V

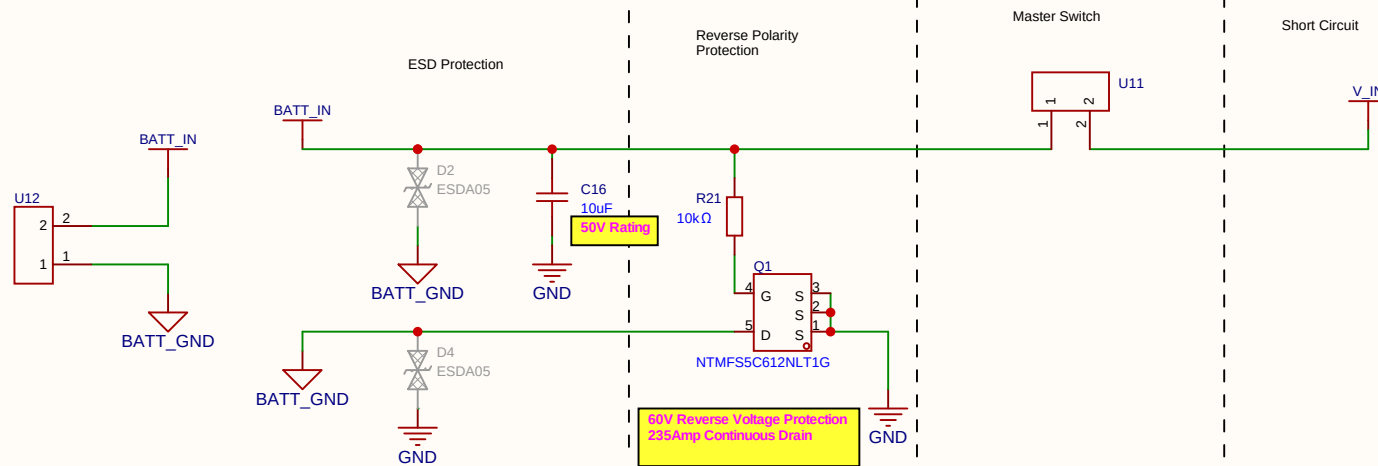


Spare output header for motor pins
Protected by 220Ωhms

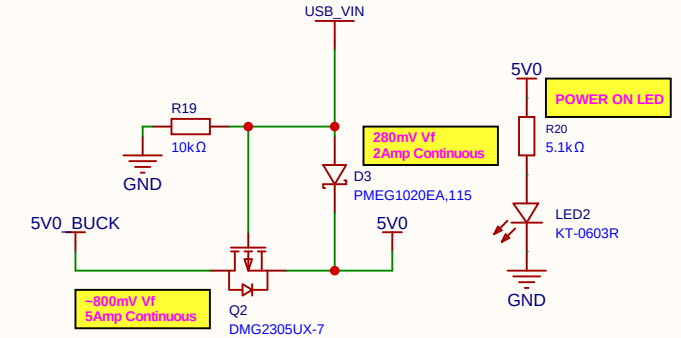


Schematic	Schematic1_1			Update Date	2025-07-31
				Create Date	2025-07-20
Page	Motor Drivers			Part Number	JLCPCB-002
Drawn	EasyEDA	Generic_Robot_Controller_ver2			
Reviewed	EasyEDA				
		VER	SIZE	PAGE	3 OF 5
EasyEDA		V0.1	A4	EasyEDA.com	

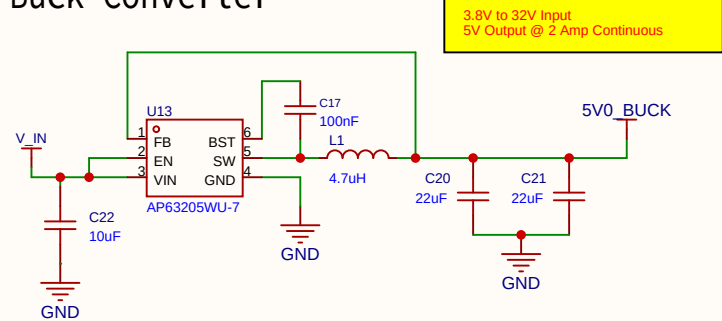
Power Protection



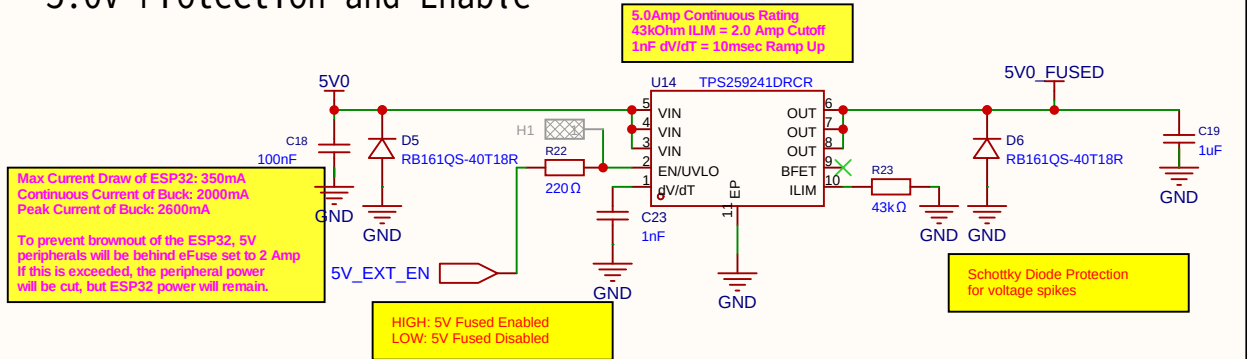
USB Power Muxing
Protect backfeeding Buck converter from USB 5V.
Will use USB power even if battery is connected.



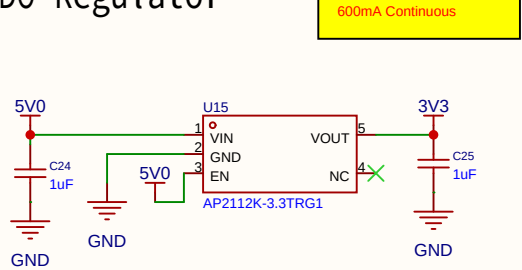
5.0V Buck Converter



5.0V Protection and Enable



3.3V LDO Regulator



Schematic	Schematic1_1			Update Date	2025-08-07
Page	Power Input and Protection			Create Date	2025-02-17
Drawn	EasyEDA	Generic_Robot_Controller_ver2			
Reviewed	EasyEDA				
		VER	SIZE	PAGE	4 OF 5
EasyEDA		V0.1	A4	EasyEDA.com	

Max Current Draw of ESP32: 350mA
Continuous Current of AP63205: 2000mA
Peak Current of AP63205: 2500mA

To prevent brownout of the ESP32, 5V peripherals will be on a 1500mA PTC Fuse. If this is exceeded, the peripheral power will be cut, but ESP32 power will remain.

4.2Amp Continuous

5V0

F1
BSMD0603L-150

PFET_S2

5V0_FUSED

5V0

R27
10kΩ

PFET_G2

5V_EXT_EN

R28
220Ω

NFET_G1

NFET_S2

NFET_G2

Q5
AP3003

1

2

3

4

5

6

PFET_G2

5V0_FUSED

GND

HIGH: 5V Fused Enabled
LOW: 5V Fused Disabled

Schematic	Schematic1_1			Update Date	2025-07-31
				Create Date	2025-07-23
Page	Removed Systems			Part Number	JLPCB-002
Drawn	EasyEDA	Generic_Robot_Controller_ver2			
Reviewed	EasyEDA				
		VER	SIZE	PAGE	5 OF 5
		V0.1	A4	EasyEDA.com	